

QUESTION 1 (Begin a new sheet)

Marks

(a) Express  $\frac{2}{4 + \sqrt{3}}$  with a rational denominator.

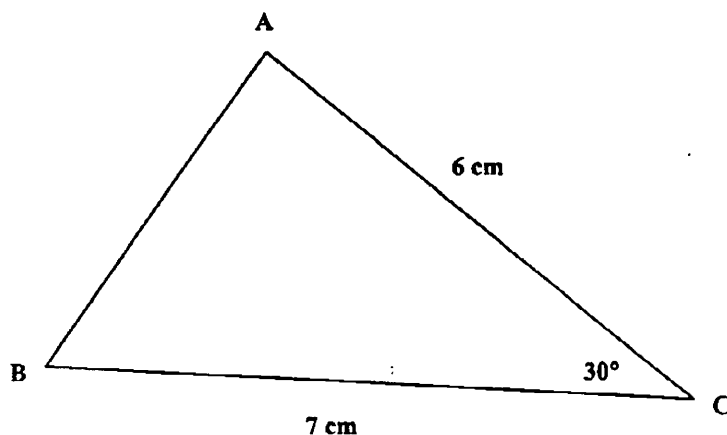
1

(b) Find the values of  $x$  for which  $6x^2 - x - 2 = 0$ .

2

(c)

2

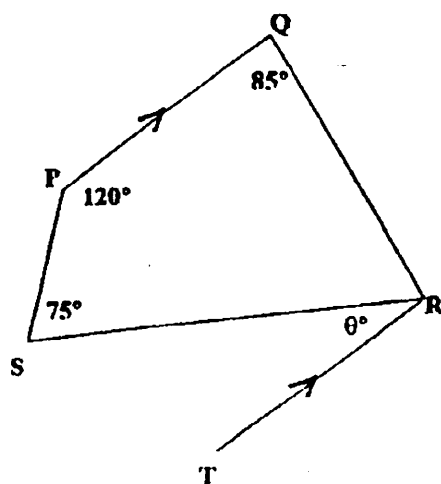


NOT TO SCALE

In the diagram,  $AC = 6$  cm,  $BC = 7$  cm and  $\angle ACB = 30^\circ$ . Use the cosine rule to find the length of  $AB$  correct to the nearest cm.

(d)

2



NOT TO SCALE

In the diagram,  $PQ \parallel TR$ ,  
 $\angle PQR = 85^\circ$ ,  $\angle QPS = 120^\circ$ ,  
 $\angle PSR = 75^\circ$  and  $\angle SRT = \theta^\circ$ .

Copy the diagram onto your answer sheet.

Find the value of  $\theta$ .

(e) Find the values of  $y$  for which  $|9y - 11| > 7$ .

2

(f) Find the primitive function of  $\sec^2 4x$ .

1

(g) A country property increased in value by  $12\frac{1}{2}\%$  to a new value of \$36 000. What was the value of the property before the increase?

2

**QUESTION 2**      (*Begin a new sheet*)

**Marks**

(a) Differentiate the following functions:

**6**

(i)  $\sqrt{3x^2 + 2}$

(ii)  $(x + 1) \ln x$

(iii)  $\frac{x}{\sin 2x}$

(b) Find:

**3**

(i)  $\int \left( x - \frac{2}{x^3} \right) dx$

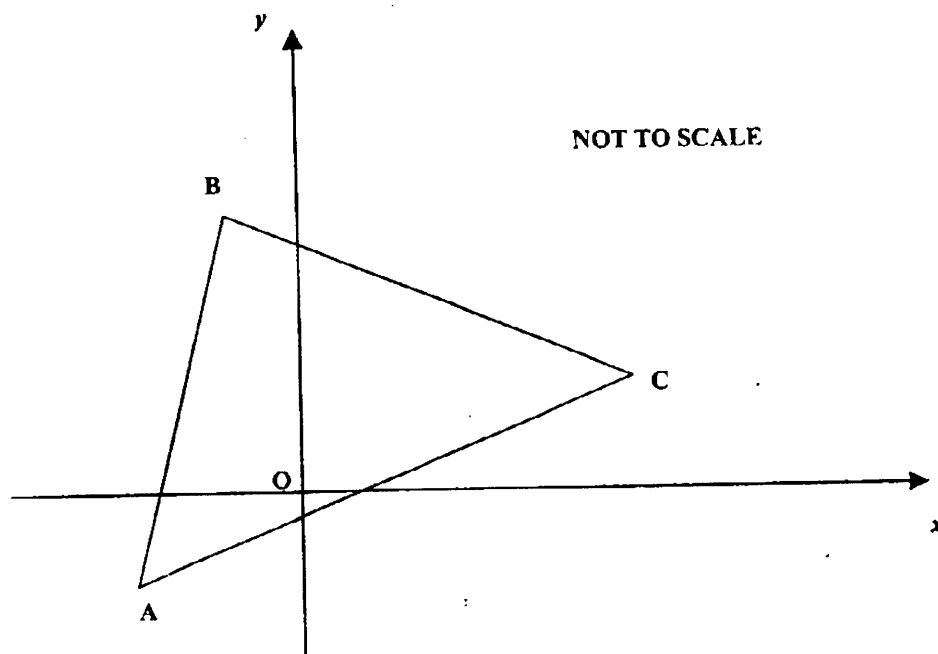
(ii)  $\int e^{3x+2} dx$

(c) Find the exact value of  $\int_0^{\pi/2} \cos \frac{x}{2} dx$

**3**

**QUESTION 3** (Begin a new sheet)

**Marks**



The diagram shows points  $A(-3,-2)$ ,  $B(-1,4)$  and  $C(5,2)$  in the Cartesian plane.

Copy this diagram onto your answer sheet.

- |                                                                                                                  |   |
|------------------------------------------------------------------------------------------------------------------|---|
| (a) Find the gradient of AC.                                                                                     | 1 |
| (b) Point P is the midpoint of AC. Show that the coordinates of P are $(1,0)$ .<br>Mark point P on your diagram. | 1 |
| (c) Show that the equation of the line perpendicular to AC and passing through the point P is $2x + y - 2 = 0$ . | 2 |
| (d) Show that B lies on the line $2x + y - 2 = 0$ .                                                              | 1 |
| (e) Show that the length of BP is $2\sqrt{5}$ units.                                                             | 1 |
| (f) Point P is the midpoint of the interval BD.                                                                  | 2 |
| (i) On your diagram show the position of point D.                                                                |   |
| (ii) Find the coordinates of D.                                                                                  |   |
| (g) Explain why the quadrilateral ABCD is a rhombus.                                                             | 1 |
| (h) Find the area of $\triangle BPC$ .                                                                           | 2 |
| (i) Hence find the area of the rhombus ABCD.                                                                     | 1 |

**QUESTION 4** (Begin a new sheet)

**Marks**

- (a) The graph of  $y = f(x)$  passes through the point  $(-1, 4)$  and  $f'(x) = 5 - 3x^2$ . Find  $f(x)$ . 2

- (b) The following table gives five values of the function  $y = f(x)$ . 2

| $x$    | 0 | 1   | 2    | 3    | 4    |
|--------|---|-----|------|------|------|
| $f(x)$ | 1 | 0.5 | 0.41 | 0.37 | 0.33 |

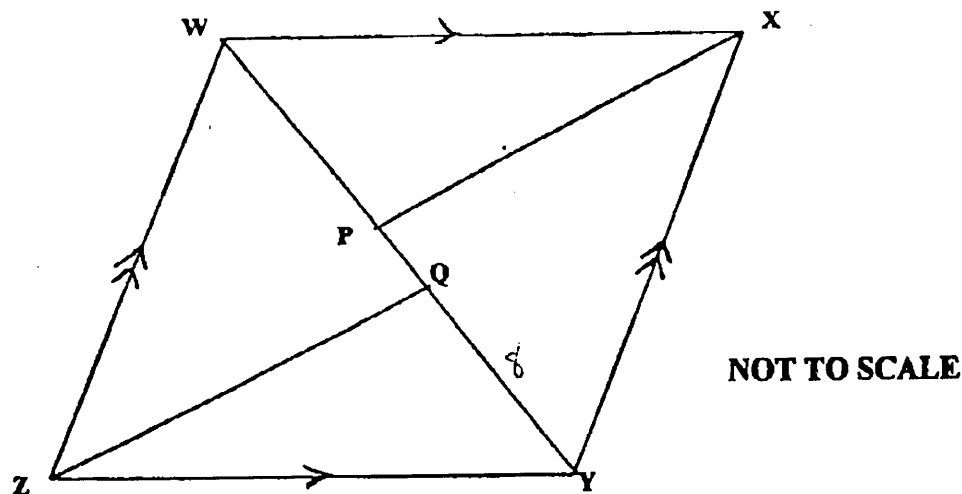
Use the five function values and Simpson's rule to approximate  $\int_0^4 f(x) dx$ .

(Give your answer correct to 2 decimal places.)

- (c) The equation of a parabola is  $(x - 3)^2 = -12(y - 1)$ . Find the: 3

- (i) coordinates of its vertex.
- (ii) equation of its directrix.

- (d) 5



WXYZ is a parallelogram. XP bisects  $\angle WXY$  and ZQ bisects  $\angle WZY$ .

Copy the diagram onto your answer sheet.

- (i) Explain why  $\angle WXY = \angle WZY$ .
- (ii) Prove  $\triangle WXP$  is congruent to  $\triangle YZQ$ .
- (iii) Hence find the length of PQ given  $WY = 20$  cm and  $QY = 8$  cm.

QUESTION 5

(Begin a new sheet)

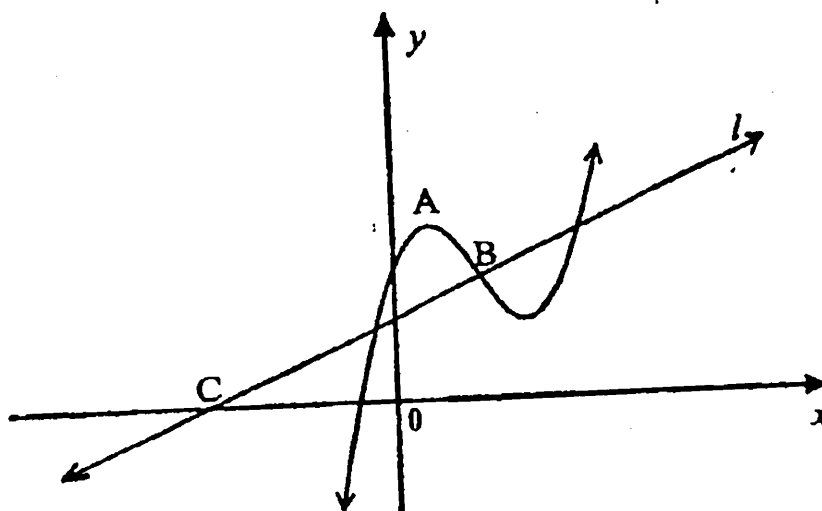
Marks

3

- (a) (i) Write down the discriminant of  $x^2 - 2kx + 6k$ .  
 (ii) For what values of  $k$  is  $x^2 - 2kx + 6k$  always positive?

9

(b)



NOT TO SCALE

The diagram shows a sketch of the curve  $y = x^3 - 6x^2 + 9x + 4$ . The curve has a local maximum point at A and a point of inflexion at B. The line  $l$  is a normal to the curve at point B and meets the  $x$  axis at point C.

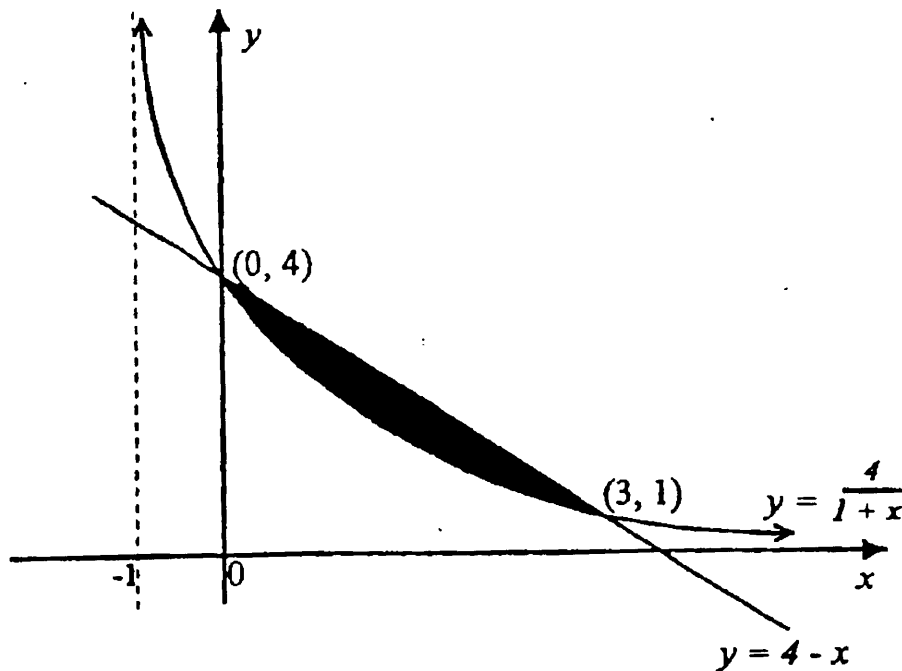
- (i) Find the coordinates of point A.  
 (ii) Show that the coordinates of point B is (2,6).  
 (iii) Show that the equation of the line  $l$  is  $x - 3y + 16 = 0$ .  
 (iv) Find the coordinates of point C.

**QUESTION 6** (Begin a new sheet)

**Marks**

(a)

3



**NOT TO SCALE**

The diagram shows part of the hyperbola  $y = \frac{4}{1+x}$  and the line  $y = 4 - x$ .

The hyperbola and line intersect at the points  $(0, 4)$  and  $(3, 1)$ .

Calculate the exact area of the shaded region.

- (b) (i) If  $x^0$  is an acute angle, find the value of  $x$  if  $3 \cos(2x) = 1$ .  
(Answer correct to 2dp)

4

- (ii) Sketch the curve  $y = 3 \cos(2x) - 1$  for the domain  $-\frac{\pi}{2} \leq x \leq \frac{\pi}{2}$  clearly showing the x and y axis intercepts and the range.

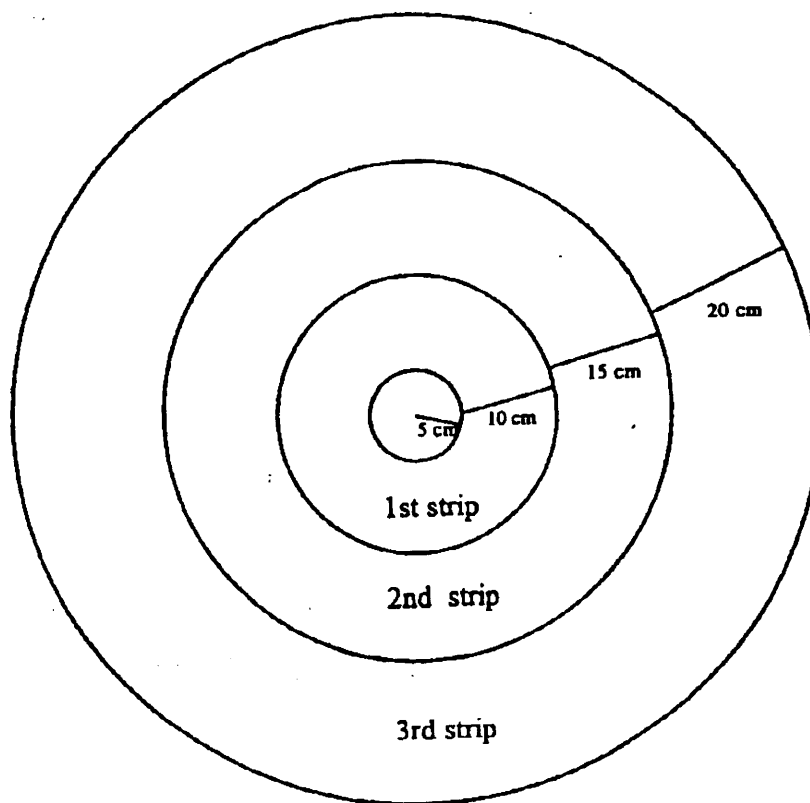
QUESTION 6

(Continued)

Marks

5

(c)



NOT TO SCALE

Beginning with a circular piece of fabric of radius 5 cm, Le sewed together circular strips of different coloured fabrics which increased in width to make a circular tablecloth. The finished width of the first strip was 10cm, the second was 15 cm, the third was 20 cm and so on.

- (i) Show that the width of the tenth strip was 55 cm.
- (ii) The radius of the table cloth was 455 cm. How many strips were sewn to the edge of the first circular piece?

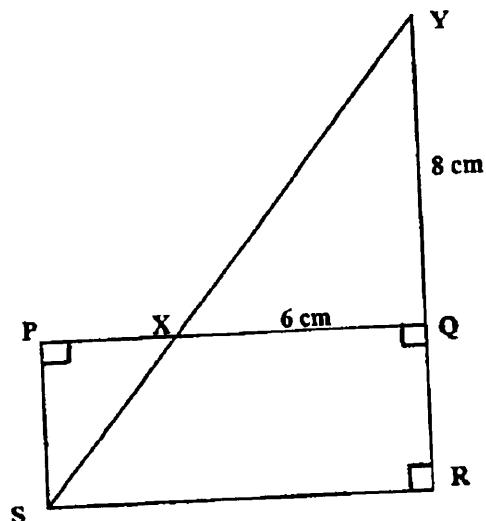
QUESTION 7

(Begin a new sheet)

Marks

3

(a)



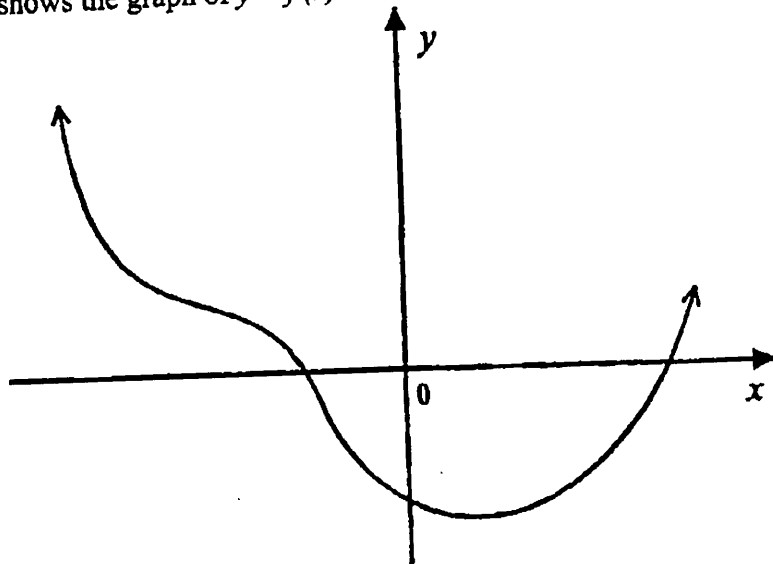
NOT TO SCALE

In the diagram, PQRS is a rectangle and  $SR = 3 PS$ . R, Q and Y are collinear points.  $XQ = 6$  cm and  $YQ = 8$  cm.

- (i) Prove  $\triangle PXS$  is similar to  $\triangle QXY$ .
- (ii) Hence find the length of PS.

2

(b) The graph shows the graph of  $y = f(x)$ .



- (i) Copy this graph onto your answer sheet.
- (ii) On the same set of axes, sketch the graph of its derivative,  $f'(x)$ .

QUESTION 7 (Continued)

Marks

(c) Consider the function  $y = \sin x + \cos x$  in the domain  $0 \leq x \leq 2\pi$ .

7

- (i) Find  $\frac{dy}{dx}$ .
- (ii) Find the maximum and minimum values of  $\sin x + \cos x$  in the given domain.
- (iii) Show that the curve cuts the  $x$  axis at  $x = \frac{3\pi}{4}$  and at  $x = \frac{7\pi}{4}$ .
- (iv) Hence sketch the curve of  $y = \sin x + \cos x$  in the domain  $0 \leq x \leq 2\pi$ .

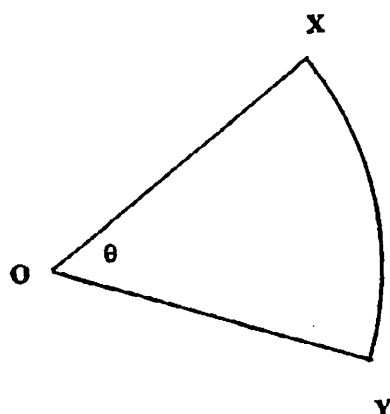
QUESTION 8

(Begin a new sheet)

Marks

2

(a)



NOT TO SCALE

In the diagram, XY is an arc of a circle with centre O and radius 12 cm. The length of the arc XY is  $4\pi$  cm. Find the:

- (i) exact size of  $\theta$  in radians.
- (ii) area of the sector OXY.

- (b) The region bounded by the curve  $y = e^x + e^{-x}$ , the x axis and the lines  $x = 0$  and  $x = 2$  is rotated about the x axis. Find the volume of the solid formed. (Leave your answer in terms of  $e$ .)

3

- (c) A particle moves along a straight line about a fixed point O so that its acceleration,  $a \text{ ms}^{-2}$ , at time  $t$  seconds is given by  $a = 8 \cos(2t + \frac{\pi}{6})$ . Initially the particle is moving to the right with a velocity of  $2 \text{ ms}^{-1}$  from a position  $\sqrt{3}$  metres to the left of O.

7

- (i) Find expressions for the velocity and position of the particle at any time  $t$ .
- (ii) Show that the particle changes directions when  $t = \frac{5\pi}{12}$  seconds.
- (iii) At what time does the particle return to its initial position for the first time?

QUESTION 9

(Begin a new sheet)

Marks

4

- (a) A super bouncy ball is dropped from a window 15m above the ground. It rebounds  $\frac{4}{5}$  of its previous height with every bounce and continues to do so until it stops. What is the total distance travelled by the ball?

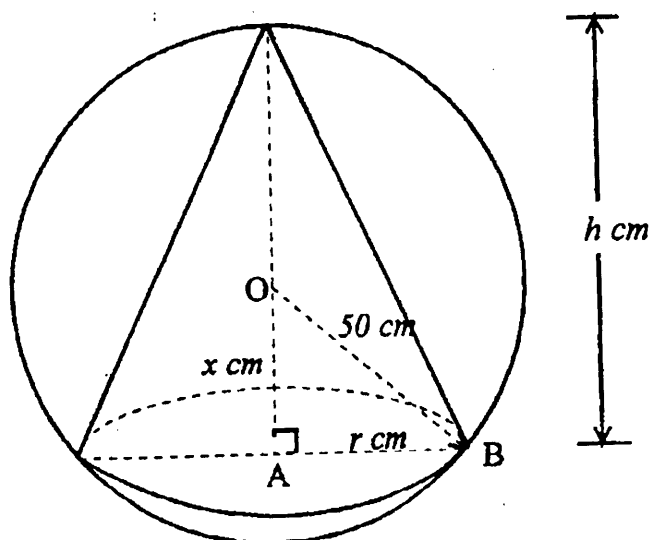
- (b) The median house price, \$P, in a certain suburb is falling at an increasing rate after a recent peak.

2

What does this tell you about  $\frac{dP}{dt}$  and  $\frac{d^2P}{dt^2}$ ?

- (c)

6



NOT TO SCALE

The diagram shows a cone of base radius  $r$  cm and height  $h$  cm inscribed in a sphere of radius 50 cm. The centre of the sphere is O and  $\angle OAB = 90^\circ$ .

Let  $OA = x$  cm.

- (i) Show that  $r = \sqrt{2500 - x^2}$ .
- (ii) Hence show that the volume,  $V$  cm<sup>3</sup>, of the cone is given by:

$$V = \frac{\pi}{3} (2500 - x^2) (50 + x)$$

- (iii) Find the radius of the largest cone which can be inscribed in the sphere. (Give your answer to the nearest mm.)

QUESTION 10 (Begin a new sheet)

Marks

- (a) In a fish hatchery, the fish population,  $N$ , satisfies the equation  $N = N_0 e^{kt}$  where  $N_0$  and  $k$  are constants and  $t$  is measured in months. 6
- (i) Initially there were 1 000 fish in the hatchery and at the end of 5 months there were 5000. Find the value of  $k$  correct to 3 decimal places.
  - (ii) Find the number of fish in the hatchery at the end of 8 months. (Give your answer correct to the nearest hundred.)
  - (iii) At the end of which month will the fish population exceed 50 000 for the first time?
  - (iv) At what rate is the population increasing at the end of six months? (Give your answer correct to the nearest hundred fish per month.)
- (b) Mario and Fei Lin worked out that they would save \$50 000 in five years by depositing all their combined monthly salary of  $\$S$  at the beginning of each month into a savings account and withdrawing \$1 600 at the end of each month for living expenses. The savings account paid interest at the rate of 3% p.a. compounding monthly. 6
- (i) Show that at the end of the second month, the balance in their savings account, immediately after making their withdrawal of \$1 600, would be given by  $\$[(1.0025^2 + 1.0025)S - 1\,600(1.0025 + 1)]$ .
  - (ii) Hence write down an expression for the balance in their account at the end of the sixtieth month.
  - (iii) Calculate their combined monthly salary.

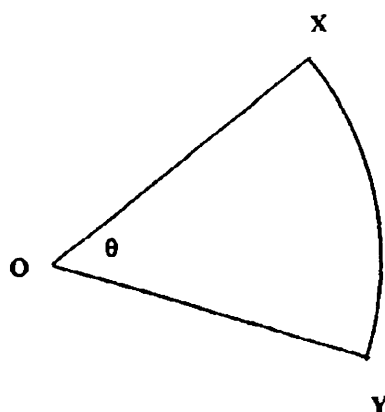
QUESTION 8

(Begin a new sheet)

Marks

(a)

2



NOT TO SCALE

$$l = r\theta$$

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3

- (c) Show that  $\frac{10}{4x^2 - 25} = \frac{1}{2x - 5} - \frac{1}{2x + 5}$

3

Hence ~~find~~ <sup>find</sup>  $\int \frac{dx}{4x^2 - 25}$

- (d) Sketch the curves  $y = \cos 2x$  and  $y = \sin \frac{1}{2}x$  in the domain  $0 \leq x \leq 2\pi$ .

4

From your sketch find the approximate solution(s) of the equation  $\cos 2x - \sin \frac{1}{2}x = 0$  for the domain  $0 \leq x \leq 2\pi$ .

**QUESTION 9***(Begin a new sheet)***Marks**

- (a) Evaluate :  $\sum_{n=3}^8 (2 \times 3^n - 2n)$ . 3
- (b) Solve :  $\log_{27} 16 = x \log_3 2$  3
- (c) Mario and Fei Lin worked out that they would save \$50 000 in five years by depositing all their combined monthly salary of \$S at the beginning of each month into a savings account and withdrawing \$1 600 at the end of each month for living expenses. The savings account paid interest at the rate of 3% p.a. compounding monthly. 6
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QUESTION 10 (Begin a new sheet)

Marks

- (a) A die numbered 1 to 6 is rolled twice. The sum  $S$  of the numbers which appear uppermost on the die is calculated.

4

- (i) Find the probability that  $S$  is greater than 9.
- (ii) It is known that 5 appears on the die at least once in the two throws. Find the probability that  $S$  is greater than 9.

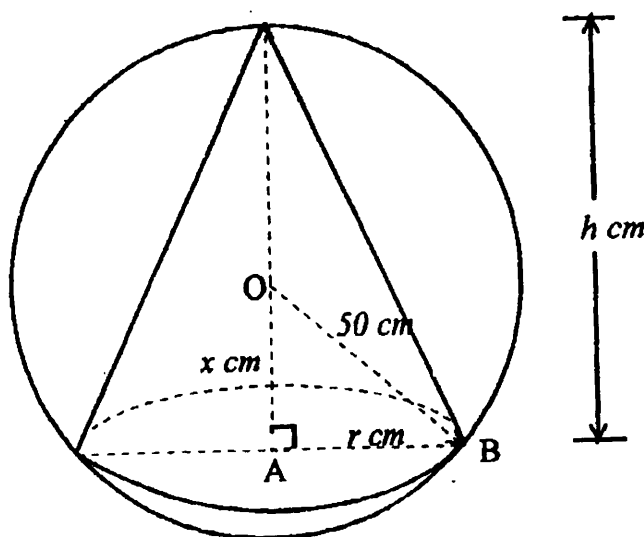
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- (iii) Find the radius of the largest cone which can be inscribed in the sphere. (Give your answer to the nearest mm.)

QUESTION 1

a)  $\frac{2}{4+\sqrt{3}} \cdot \frac{4-\sqrt{3}}{4-\sqrt{3}} = \frac{2(4-\sqrt{3})}{16-3}$   
 $= \frac{2(4-\sqrt{3})}{13}$  (1m)

b)  $6x^2 - x - 2 = 0$   
 $(3x-2)(2x+1) = 0$   
 $x = -\frac{1}{2}, \frac{2}{3}$  (2m)

c)  $c^2 = a^2 + b^2 - 2ab \cos C$   
 $AB^2 = 7^2 + 6^2 - 2 \times 7 \times 6 \cos 30^\circ$  (2m)  
 $AB = 3.5005 \dots$   
 $\approx 4 \text{ cm}$

d)  $\angle QRS = 80^\circ$  (Angle sum of Quadrilateral)

$\angle QRT = 95^\circ$  (Co-interior Angles) (1m)

$\angle SRT = \angle QRT - \angle QRS$  (Using Adjacent Angles)  
 $= 95^\circ - 80^\circ$   
 $= 15^\circ$  (2m)

e)  $|9y - 11| > 7$

$9y - 11 > 7$  or  $9y - 11 < -7$   
 $9y > 18$  or  $9y < 4$   
 $y > 2$  or  $y < \frac{4}{9}$  (2m)

f)  $\int \sec^2 4x dx = \frac{1}{4} \tan 4x + C$  (1m)  
 ignore C

g)  $112\frac{1}{2}\%$  of  $x = \$36000$ ,  $x$  is original price  
 $x = \$32000$  (2m)  
 $\approx \$32000$

QUESTION 2

a) (i)  $\frac{d}{dx} \{(3x^2+2)^{\frac{1}{2}}\} = \frac{1}{2}(3x^2+2)^{-\frac{1}{2}} \cdot 6x$  (2m)  
 $= 3x(3x^2+2)^{-\frac{1}{2}}$   
 $= \frac{3x}{\sqrt{3x^2+2}}$

(ii)  $\frac{d}{dx} \{(x+1) \ln x\} = (x+1) \cdot \frac{1}{x} + (\ln x) \cdot 1$  (2m)  
 $= \frac{x+1}{x} + \ln x$

(iii)  $\frac{d}{dx} \left\{ \frac{x}{\sin 2x} \right\} = \frac{1 \cdot \sin 2x - (2 \cos 2x) x}{\sin^2 2x}$  (2m)  
 $= \frac{\sin 2x - 2x \cos 2x}{\sin^2 2x}$

b) (i)  $\int (x - 2x^3) dx = \frac{1}{2}x^2 - \frac{2}{4}x^4 + C$   
 $= \frac{1}{2}x^2 - \frac{1}{2}x^4 + C$  (2m)  
 $= \frac{x^4 + 2}{2x^2} + C$

(ii)  $\int e^{3x+2} dx = \frac{1}{3} e^{3x+2} + C$  (1m)

c)  $I = \left[ 2 \sin \frac{x}{2} \right]_0^{\frac{\pi}{2}}$  (1m)  
 $= 2 \left[ \sin \frac{\pi}{4} - \sin 0 \right]$  (3m)  
 $= 2 \left[ \frac{\sqrt{2}}{2} - 0 \right] = \sqrt{2}$

QUESTION 3

a)  $\frac{1}{2}$  (1m)

b)  $P\left(-\frac{3+5}{2}, \frac{-2+2}{2}\right) = P(1, 0)$  (1m)

c)  $m = -2; (1, 0)$   $y - 0 = -2(x - 1)$  (2m)  
 $y = -2x + 2$   
 $2x + y - 2 = 0$

d) L.H.S.  $= 2(-1) + (4) - 2 = 0 = \text{R.H.S.}$  (1m)

e)  $BP = \sqrt{(1+1)^2 + (0-4)^2} = \sqrt{20} = 2\sqrt{5}$  (1m)

f) (i)  $(1, 0)$  (1m) (ii)  $(-1, 4)$  (1, 0) D(x, y)  
 $(1, 0) = \left( \frac{-1+x_2}{2}, \frac{4+y_2}{2} \right)$   
 $x_2 = 3, y_2 = -4$  (1m)

(1m) g) AC, BD DIAGONALS BISECT AT  $90^\circ$

h)  $A = \frac{1}{2} \times \sqrt{5} \times 2\sqrt{5} = 10$  (2m) i) 4 SQUARES UNITS (1m)

Q4

a)  $f'(x) = 5 - 3x^2$

$f(x) = 5x - x^3 + C$

$4 = -5 + 1 + C$

$\therefore C = 8$

$f(x) = 5x - x^3 + 8$

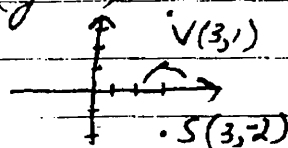
b)  $\int_0^4 f(x) dx = \frac{1}{3} \{ (0.33+1) + 4(0.5+0.57) + 2(0.4) \}$   
 $\approx 1.88$

c)  $(x-3)^2 = -12(y-1)$

(i)  $V(3,1)$

(ii)  $y = 4$

using  $a=3$



d) (i) opposite angles of parallelogram are equal as

$\angle W + \angle X = 180^\circ$  (CO-INTERIOR ANGLES)

$\angle W + \angle Z = 180^\circ$  ( " )

$\therefore \angle X = \angle Z$  in  $\triangle WXY = \triangle WZY$

(ii)  $WX = ZX$  (opp sides of parallelogram)

$\angle XWP = \angle ZYP$  (alternate angles)

$\angle WXP = \angle ZYP$  (each is half of  $\angle X$  and  $\angle Z$ )

$\therefore \triangle WXP \cong \triangle ZYP$  (AAS TEST)

(iii)  $WP = YP$  (corresp sides of congruent triangles)

$WP + PQ + PY = 20$

$x + PQ + x = 20$

$8 + PQ + 8 = 20$

$\therefore PQ = 4$

Q5

a)  $\Delta = b^2 - 4ac$   
 $= (-2k)^2 - 4 \times 1 \times 6k$   
 $= 4k^2 - 24k$

(i)  $\Delta < 0$

$4k^2 - 24k < 0$

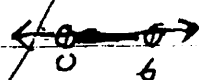
$k^2 - 6k < 0$

$k(k-6) < 0$

$0 < k < 6$

1m

2m



b)  $y = x^3 - 6x^2 + 9x + 4$

(i)  $\frac{dy}{dx} = 3x^2 - 12x + 9$

Put  $\frac{dy}{dx} = 0$

$3x^2 - 12x + 9 = 0$

$x^2 - 4x + 3 = 0$

$(x-1)(x-3) = 0$

$x = 1, 3$

$(1, 8) \quad (3, 4)$

Test  $(1, 8)$

$\frac{d^2y}{dx^2} = 6x - 12$  (Test)

At  $x=1, \frac{d^2y}{dx^2} = -6 < 0 \Rightarrow$  MAX

(ii)  $6x - 12 = 0$  in  $y'' = 0$

$x = 2 \therefore y = 6$

Test either side

$f''(1.5) = -3 < 0 \Rightarrow$  CHANGE OF

$f''(2.5) = 3 > 0 \Rightarrow$  CONCAVITY

(iii) At  $B(3, 6)$

$\frac{dy}{dx} = 3(2^2 - 4 \times 2 + 3) = -3$

$\therefore$  Normal has grad  $\frac{1}{3}$

Equation is

$y - 6 = \frac{1}{3}(x - 3)$

$3y - 18 = x - 3$

$x - 3y + 15 = 0$  (L)

iv) Put  $y = 0$  in (L)

$x + 15 = 0$

$x = -15$  in  $(-15, 0)$

1m

Q6

$$a) A = \int_0^3 \left\{ (4-x) - \left( \frac{4}{1+x} \right) \right\} dx$$

$$= \left[ 4x - \frac{x^2}{2} - 4 \ln|1+x| \right]_0^3$$

$$= \left[ \left( 12 - \frac{9}{2} - 4 \ln 4 \right) - (0 - 4 \ln 1) \right]$$

$$= 7\frac{1}{2} - 4 \ln 4$$

$$\therefore \left( \frac{15}{2} - 4 \ln 4 \right) \text{ square units}$$

b) 0.7/Yes  $\begin{matrix} 0.7 & Y & 0.7 & Y \\ 0.3 & Y & 0.3 & N \\ & N & 0.7 & Y \\ & & 0.3 & N \end{matrix}$

$$(i) P(YYY) = (0.7)^3 = 0.343 \quad (1m)$$

$$(ii) P(X \geq 1) = 1 - P(X=0) = 1 - P(NNN) = 1 - (0.3)^3 = 0.973 \quad (2m)$$

c) This is arithmetic series

$$(i) a=10, d=5$$

$$T_{10} = a + 9d = 10 + 9 \times 5 = 55 \quad (2m)$$

$$(ii) S_n = 450$$

$$\frac{n}{2} (2a + (n-1)d) = 450$$

$$\frac{n}{2} (20 + (n-1)5) = 450$$

$$\frac{n}{2} (15 + 5n) = 450$$

$$\frac{n}{2} (3+n) = 90$$

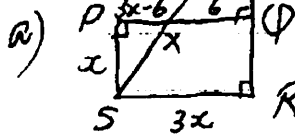
$$n^2 + 3n - 180 = 0$$

$$(n+15)(n-12) = 0$$

$$n = 12$$

12 strips

Q7



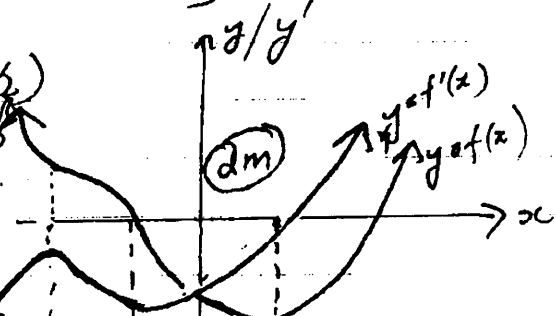
(i) In  $\Delta PXS, \Delta QXY$   
 $\angle SPX = \angle YQX = 90^\circ$  (Given)  
 $\angle PXS = \angle QXY$  (Vert opposite angle)  
 $\therefore \Delta PXS \parallel \Delta QXY$  (equiangular)

$$(ii) \frac{8}{x} = \frac{6}{3x-6} \quad (\text{Ratios of corresponding sides})$$

$$6x = 24x - 48$$

$$x = \frac{8}{3} \text{ i.e. } PS = \frac{8}{3} \text{ cm} \quad (1m)$$

(1m) OR ALTERNATIVE TECHNIQUE



$$c) (i) y = \sin x + \cos x$$

$$\frac{dy}{dx} = \cos x - \sin x \quad (1m)$$

$$(ii) \text{ Put } \frac{dy}{dx} = 0$$

$$\sin x = \cos x$$

$$\therefore \tan x = 1$$

$$\therefore x = \frac{\pi}{4}, \frac{5\pi}{4} \quad (2m)$$

$$y'' = -\sin x - \cos x$$

$$= -(\sin x + \cos x) \quad (TESTS)$$

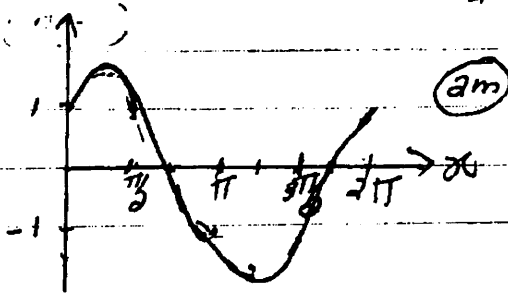
$$\text{at } x = \frac{\pi}{4}, y'' = -\sqrt{2} < 0 \Rightarrow \text{MAX TURN PT}$$

$$\text{at } x = \frac{5\pi}{4}, y'' = \sqrt{2} > 0 \Rightarrow \text{MIN TURN PT}$$

$$(iii) y = 0 \text{ i.e. } \sin x + \cos x = 0$$

$$\tan x = -1$$

$$x = \frac{3\pi}{4}, \frac{7\pi}{4} \quad (2m)$$



08

a)  $l = r\theta$

$4\pi = 2\theta$

$\theta = \frac{\pi}{3}$  (1m)

(10)  $A = \frac{1}{2}r^2\theta$

$= \frac{1}{2} \times 12^2 \times \frac{\pi}{3}$

$= 24\pi$

$\approx 24\pi \text{ cm}^2$  (1m)



$V = \pi \int_0^2 (e^x + e^{-x})^2 dx$

$= \pi \int_0^2 (e^{2x} + 2 + e^{-2x}) dx$

$= \pi \left[ \frac{e^{2x}}{2} + 2x - \frac{e^{-2x}}{2} \right]_0^2$

$= \pi \left[ \left( \frac{e^4}{2} + 4 - \frac{1}{2e^4} \right) - \left( \frac{1}{2} + 0 - \frac{1}{2} \right) \right]$

$= \pi \left( \frac{e^8 + 8e^4 - 1}{2e^4} \right)$

ie  $\pi \left( \frac{e^8 + 8e^4 - 1}{2e^4} \right)$  CUBIC UNITS

c)  $\frac{1}{2x-5} - \frac{1}{2x+5} = \frac{2x+5 - (2x-5)}{(2x+5)(2x-5)}$

$= \frac{10}{(2x+5)(2x-5)} = \frac{10}{4x^2 - 25}$

$\frac{dx}{4x^2 - 25} = \frac{1}{10} \left[ \int \frac{1}{2x-5} - \frac{1}{2x+5} \right] + C$

$= \frac{1}{10} \ln \left( \frac{2x-5}{2x+5} \right) + C$

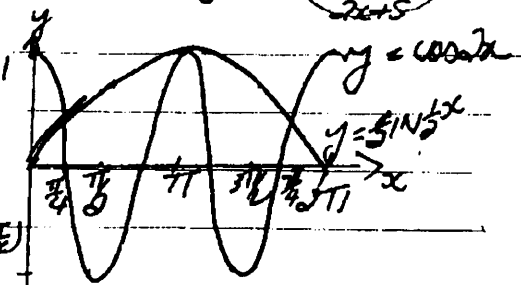
d)

(2m) for sketch

$x = \frac{\pi}{5}, \pi, \frac{9\pi}{5}$

$(\frac{\pi}{2})$   $(\frac{7\pi}{2})$

(2m)



09

a)  $S = (2 \times 3^3 - 6) + (2 \times 3^4 - 8) + (2 \times 3^5 - 10) + \dots + (2 \times 3^9 - 16)$

$= 2(3^3 + 3^4 + \dots + 3^9) - (6 + 8 + 10 + \dots + 16)$

$= 2 \cdot 3^3(3^6 - 1) - \frac{1}{2}(6 + 16) \cdot 2$

$= 3^3(3^6 - 1) - 66$

$= 19590$  (3m)

b)  $\log_2 16 = x \log_3 2$

$x \log_3 2 = \frac{\log_3 16}{\log_3 2}$

$x \log_3 2 = \log_3 16$

$x = \frac{4 \log_3 2}{3 \log_3 2} = \frac{4}{3}$  (3m)

[SEE BACK PAGE]

c) Let  $A_t$  be accumulated money at end of month after withdrawal

$A_1 = (1.0025)S - 1600$

$A_2 = [(1.0025)S - 1600 + S] \cdot 1.0025$

$A_3 = [(1.0025)^2 S - 1600(1 + 1.0025) + S] \cdot 1.0025$

$A_{60} = [1.0025^{60} S - 1600(1.0025^{60} - 1.0025^0) + S] \cdot 1.0025^{60}$

$= 1600[1.0025^{60} - 1] + 50000 \cdot 1.0025^{60}$

SINCE  $A_{60} = 50000$

$50000 = 1600[1.0025^{60} - 1] + 50000 \cdot 1.0025^{60}$

$S = \frac{50000 + 1600[1.0025^{60} - 1]}{1.0025^{60} - 1}$

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$S = \frac{50000 + 1600[1.0025^{60} - 1]}{1.0025^{60} - 1}$

Q 10

$$S = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

$$f = \{1, 2, 3, 4, 5, 6, 5, 4, 3, 2, 1\}$$

a(i)

$$E = \{(5,6), (6,4), (5,5), (5,6), (6,5), (6,6)\}$$

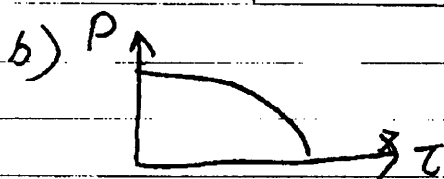
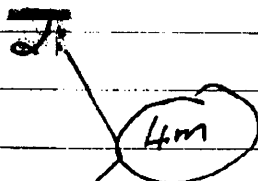
$$P(E) = \frac{6}{36} = \frac{1}{6}$$

(ii)

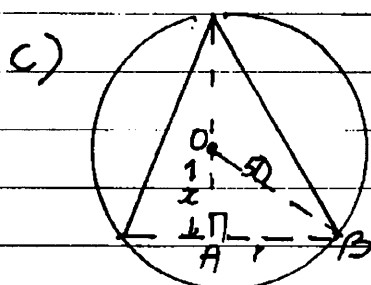
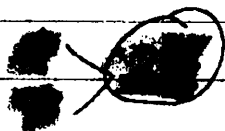
$$\begin{array}{|c|c|} \hline 5 & 5 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 5 & 5 \\ \hline \end{array} \quad \begin{array}{|c|c|} \hline 5 & 5 \\ \hline \end{array}$$

$$5-6 \quad 6-5 \quad 5-5$$

$$P(E_3) = \frac{3}{36} = \frac{1}{12}$$



$$\frac{dP}{dx} < 0$$



From Pythagoras

(i)

$$r^2 + x^2 = 50^2$$

$$r = \sqrt{2500 - x^2}$$

h)

$$V = \frac{1}{3} \pi (2500 - x^2)(50 + x)$$

$$V = \frac{1}{3} \pi (2500 - x^2)(50 + x)$$

(iii)

$$\frac{dV}{dx} = \frac{1}{3} \pi \{ (2500 - x^2) \cdot 1 + (50 + x) \cdot (-2x) \}$$

$$= \frac{1}{3} \pi \{ 2500 - x^2 - 100x - 2x^2 \}$$

$$= \frac{1}{3} \pi \{ 2500 - 100x - 3x^2 \}$$

Put  $\frac{dV}{dx} = 0$  for maximum

$$2500 - 100x - 3x^2 = 0$$

$$2500 - 100x - 3x^2 = 0$$

$$3x^2 + 100x - 2500 = 0$$

$$(3x - 50)(x + 50) = 0$$

$$x = -50, \frac{50}{3}$$

$$\frac{d^2V}{dx^2} = \frac{1}{3} \pi \{ -100 - 6x \}$$

At  $x = \frac{50}{3}$ ,  $\frac{d^2V}{dx^2} < 0 \Rightarrow \text{MAX}$

$\therefore r = \sqrt{2500 - \frac{2500}{9}}$  ie

Q9 c)

(i) Let  $A_1$  be the amount accumulated after 1 month and after withdrawal

$$A_1 = (1.0025)S - 1600$$

$$A_2 = [(1.0025)S - 1600 + S] 1.0025 - 1600$$

$$\therefore = [(1.0025)^2 + 1.0025]S - 1600[1 + 1.0025]$$

$$(ii) A_{60} = [1.0025 + (1.0025)^2 + \dots + (1.0025)^{59}]S - 1600[1 + 1.0025 + \dots + (1.0025)^{59}]$$

$$(iii) A_{60} = 50000$$

$$[1.0025 + (1.0025)^2 + \dots + (1.0025)^{59}]S - 1600[1 + 1.0025 + \dots + (1.0025)^{59}] = 50000$$

$$S = \frac{50000 + 1600[1 + 1.0025 + \dots + (1.0025)^{59}]}{1.0025 + (1.0025)^2 + \dots + (1.0025)^{60}}$$

$$= \frac{50000 + 1600 \left[ \frac{1.0025^{60} - 1}{0.0025} \right]}{\left[ \frac{1.0025(1.0025^{60} - 1)}{0.0025} \right]}$$

$$= 2325.73$$

(2m)

(1m)

1m

1m

1m

(3m)